

Self-Similarity and Fractals

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Overview

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Motivation and Context

- Fractals: infinitely complex objects, generally defined by simple rules. Most fractals we will look at are “self-similar” (i.e., made up of smaller parts of itself).
- Mandelbrot (1975): coined the term “fractal” as a shape whose Hausdorff dimension exceeds topological dimension.
- Applications: coastlines, clouds, mountains, finance, physics, economics.
- Aim of thesis: use self-similarity to investigate fractal properties via Hausdorff metric & measure.

Mathematical Tools

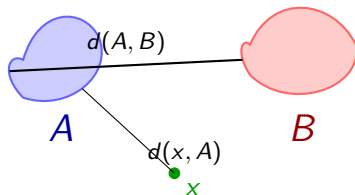
The mathematical tools we need to investigate this are:

- **Hausdorff Metric:** compares compact sets and allows us to talk about convergence of sets.
- **Hausdorff Measure & Dimension:** a generalisation of Lebesgue measure to arbitrary metric spaces. Allows us to assign sizes to sets.
- **Self-Similarity:** allows us to explore how a fractal is defined geometrically
- **Open Set Condition (OSC):** the connecting piece between self-similarity and the Hausdorff Measure, ensures minimal overlap of components of the fractal

The Hausdorff Metric

Let (X, d) be a metric space and let $A, B \subseteq X$, $A \neq \emptyset$ and $x \in X$. Recall:

$$d(x, A) := \inf_{a \in A} d(x, a), \quad d(A, B) := \sup_{a \in A} d(a, B) = \sup_{a \in A} \inf_{b \in B} d(a, b).$$



Hausdorff metric (definition)

Define

$$\mathcal{K}(X) = \{A \subseteq X : A \text{ is compact, } A \neq \emptyset\}.$$

Then

$$d_H(A, B) := \max\{d(A, B), d(B, A)\}$$

is a metric on $\mathcal{K}(X)$. This is called the induced Hausdorff metric with respect to the metric space (X, d) . Moreover $(\mathcal{K}(X), d_H)$ is complete iff (X, d) is complete.

Hausdorff measure (definition)

Let (X, d) be a metric space with $\text{diam}(\emptyset) := 0$. For $s \geq 0$, $\delta > 0$ define

$$\mathcal{H}_\delta^s(A) := \inf \left\{ \sum_{n=1}^{\infty} (\text{diam}(E_n))^s : A \subseteq \bigcup_{n=1}^{\infty} E_n, \text{diam}(E_n) \leq \delta \right\}.$$

If A does not possess any δ -coverings, we have $\mathcal{H}_\delta^d(A) := \inf \emptyset = \infty$. The Hausdorff s -measure is

$$\mathcal{H}^s(A) := \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(A) = \sup_{\delta > 0} \mathcal{H}_\delta^s(A).$$

If $d = 0$: Define $\text{diam}(\emptyset)^0 = 0$ and use the interpretation $0^0 = 1$ for any other sets.

Important properties

- It is an outer measure on X . So it is a measure on the σ -algebra of Hausdorff measurable sets
- In the Euclidean space, $(\mathbb{R}^n, d_E)$, the Hausdorff measure recovers Lebesgue measure (up to normalising constant) on for $n \in \mathbb{N}^+$. In any metric space, $\mathcal{H}^0$ is the counting measure. For $A \subseteq \mathbb{R}^n$,

$$\lambda_n(A) = \frac{\pi^{n/2}}{2^n \Gamma(n/2 + 1)} \mathcal{H}^n(A), \quad \Gamma(t) := \int_0^\infty e^{-x} x^{t-1} dx$$

- Invariant under isometries: If $f : X \rightarrow X$ is an isometry, then for any $A \subseteq X$, $\mathcal{H}^n(A) = \mathcal{H}^n(f(A))$
- Translation and scaling relationships: In metric space $(\mathbb{R}^n, d_E)$, $\mathcal{H}^n(A) = \mathcal{H}^n(A + \mathbf{x})$ and for $\lambda > 0$, $\mathcal{H}^n(\lambda A) = \lambda^n \mathcal{H}^n(A)$

Important properties

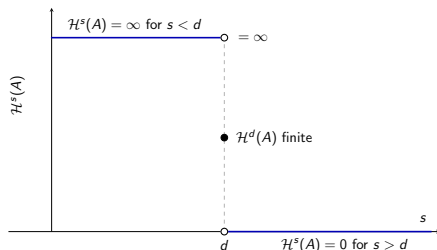
- Let (X, d) be a metric space, $A \subseteq X$, and $0 \leq s < t$. Then $\mathcal{H}^s(A) < \infty \Rightarrow \mathcal{H}^t(A) = 0$ and $\mathcal{H}^t(A) > 0 \Rightarrow \mathcal{H}^s(A) = \infty$.
 - This creates a “threshold behaviour” for any set A
 - If $\mathcal{H}^d(A)$ is a finite, nonzero value, then for every $s \in [0, \infty)$ with $s \neq d$, we either have $\mathcal{H}^s(A)$ is infinite, or zero. More specifically,

$$\mathcal{H}^s(A) = \begin{cases} \infty & \text{if } s < d \\ \text{finite, nonzero number} & \text{if } s = d \\ 0 & \text{if } s > d \end{cases}$$

- At this critical dimension, the measure value may be zero or infinite. Moreover, the dimension may be zero or infinite.

The Hausdorff Dimension

Visually, the graph looks something like:



In this graph, we will say that the dimension of set A is this critical point d . For a set $A \subseteq X$, the Hausdorff dimension is a map $\dim_H : \mathcal{P}(X) \rightarrow [0, \infty]$ defined by:

$$\dim_H(A) := \inf\{d \geq 0 : \mathcal{H}^d(A) = 0\} = \sup\{d \geq 0 : \mathcal{H}^d(A) = \infty\}$$

For this definition, we take $\sup \emptyset = 0$ and $\inf \emptyset = \infty$

Important Properties of the Hausdorff Dimension

- Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \rightarrow Y$ is bi-lipschitz if there is some real number $\alpha \geq 1$ such that

$$\frac{1}{\alpha} d_X(x, y) \leq d_Y(f(x), f(y)) \leq \alpha d_X(x, y)$$

Let $f : X \rightarrow X$ be a bi-lipschitz map and $A \subseteq X$. Then $\dim_H(A) = \dim_H(f(A))$

- In metric space $(\mathbb{R}^n, d_E)$, $\dim_H(A + \mathbf{x}) = \dim_H(A)$ and $\dim_H(\lambda A) = \dim_H(A)$
- Monotonicity: If $A \subseteq B$, then $\dim_H A \leq \dim_H B$
- Stable under countable unions: $\dim_H \left(\bigcup_{n=1}^{\infty} A_n \right) = \sup_{n \in \mathbb{N}^+} \dim_H(A_n)$
- It agrees with our usual notion of dimensions: lines are 1-dimensional, squares are 2-dimensional, cubes are 3-dimensional while also assigning dimensions to irregular sets.

Self-Similarity

A map $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called a *similitude* if there exists $r \in (0, 1)$ such that

$$|S(\mathbf{x}) - S(\mathbf{y})| = r|\mathbf{x} - \mathbf{y}|, \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

These are simply contractions in metric space $(\mathbb{R}^n, d_E)$. We call r the contraction ratio of S .

An Iterated Function System (IFS) is a finite collection $\mathcal{S} = \{S_k : \mathbb{R}^n \rightarrow \mathbb{R}^n\}_{k=1}^N$ of similitudes with $N \geq 2$, and contraction ratios $r_1, \dots, r_N$.

A non-empty compact set K is invariant under $\mathcal{S}$ if $K = \bigcup_{k=1}^N S_k(K)$

Self-Similarity Extended

- For an Iterated Function System $\mathcal{S}$, there is a unique compact set invariant under $\mathcal{S}$
- Let $\mathcal{S}$ be an IFS. If

$$\sum_{i=1}^N r_i^D = 1$$

then we call D the *similarity dimension* of $\mathcal{S}$ (denoted $\dim_{\mathcal{S}} \mathcal{S}$).

- **OSC: The Open Set Condition:** An IFS $\mathcal{S}$ satisfies the *Open Set Condition* if there exists a nonempty open set O such that

$$\bigcup_{i=1}^N S_i(O) \subseteq O \text{ and } S_i(O) \cap S_j(O) = \emptyset \text{ for } i \neq j$$

Self-Similarity Final

- Let K be the unique invariant set under $\mathcal{S} = \{S_i\}_{i=1}^N$, and $s = \dim_H K$. Then K is self-similar if

$$\mathcal{H}^s(S_i(K) \cap S_j(K)) = 0$$

for $i \neq j$ and $1 \leq i, j \leq N$, $i, j \in \mathbb{N}^+$

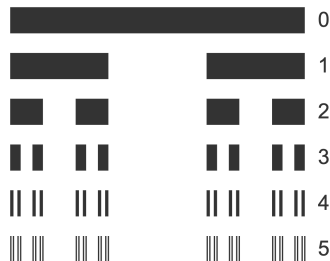
- If $\mathcal{S}$ satisfies the OSC, then the invariant set K of $\mathcal{S}$ is self-similar and $0 < \mathcal{H}^D(K) < \infty$ where D is the unique number for which

$$\sum_{i=1}^N r_i^D = 1$$

That is, $\dim_{\mathcal{S}} \mathcal{S} = \dim_H K$ (the similarity dimension is the same as the Hausdorff dimension). This gives us an easy way to calculate the dimension of self-similar sets.

The Cantor Set

- Begin with the interval $C_0 = [0, 1]$
- Remove the middle third $(\frac{1}{3}, \frac{2}{3})$ to get $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$
- Remove the middle third from each remaining interval to get $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$
- Continue this process indefinitely, with C_n the result at the n th iteration
- The Cantor set is the set as $n \rightarrow \infty$, that is, $\mathcal{C} := \lim_{n \rightarrow \infty} C_n = \bigcap_{n \in \mathbb{N}_0} C_n$



The Cantor Set

- We can alternatively define the Cantor set in terms of similitudes. Notice that if $C_0 = [0, 1]$, then $C_{n+1} = \frac{C_n}{3} \cup (\frac{C_n}{3} + \frac{2}{3})$
- So for similitudes $S_1(x) = \frac{1}{3}x$ and $S_2(x) = \frac{1}{3}x + \frac{2}{3}$, we get that

$$C = S_1(C) \cup S_2(C)$$

- So the Cantor Set C is the unique invariant set under $\{S_1, S_2\}$.
- Thus finding $d := \dim_H C = \dim_S \{S_1, S_2\}$ is the same as finding the solution to

$$2 \left(\frac{1}{3} \right)^d = 1 \iff d = \frac{\ln 2}{\ln 3}$$



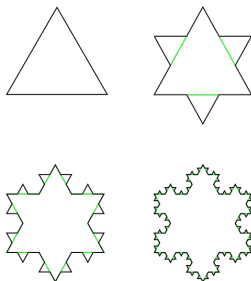
The Cantor Set Further Properties

- It is totally disconnected
- It is uncountable
- It is a set with an irrational dimension < 1
- So $\lambda_1(\mathcal{C}) = 0$
- It is self-similar and $\mathcal{H}^{\ln 2 / \ln 3}(\mathcal{C}) > 0$

The Koch Snowflake

To geometrically construct the Koch Snowflake, start with an equilateral triangle (say of side length 1). Then recursively, for each line segment,

- Divide the line segment into 3 segments of equal length
 - Draw an equilateral triangle that has the middle segment as its base and points outward
 - Remove the line segment that is the base of this triangle
- We can show that the Hausdorff dimension is $\frac{\ln 4}{\ln 3}$
 This is greater than 1 so the total length of the snowflake is ∞
 We can show that the area inside the Koch Snowflake is $\frac{2\sqrt{3}}{5}$



Sierpiński Triangle & Carpet

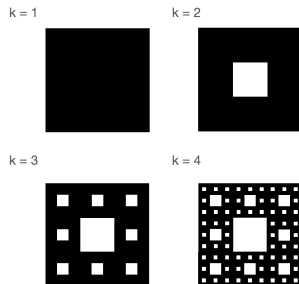
The Sierpinski Triangle:

- Start with an equilateral triangle (with its area, say side length 1).
- Subdivide the triangle into 4 congruent triangles, and remove the middle triangle.
- Repeat this recursively on all remaining triangles



The Sierpinski Carpet:

- Begin with a square (let's say $[0, 1] \times [0, 1]$).
- Subdivide this square into 9 congruent squares, and remove the middle square.
- Repeat this recursively on all remaining squares



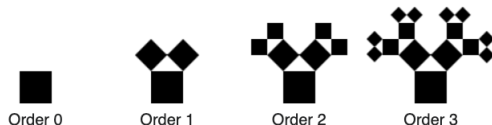
Sierpiński Triangle & Carpet properties

- Triangle dimension: $\frac{\ln 3}{\ln 2} \approx 1.585$.
- The triangle has infinite perimeter but no area
- Carpet dimension: $\frac{\ln 8}{\ln 3} \approx 1.893$.
- The carpet again has no area

Pythagoras Tree

Some fractals cannot use similitudes to describe. Instead we use the Hausdorff measure directly

- Begin with a unit square (let's say the bottom left corner is $(0,0)$).
- On the top edge of this square, construct an isosceles right triangle with the top edge as its hypotenuse. On each of the two equal sides of the triangle, construct a new square outward.
- Repeat the process recursively on every new square.
- We let $\mathcal{P}_n$ be the n -th iteration and the Pythagoras' Tree be $\mathcal{P} = \bigcup_{n \in \mathbb{N}_0} \mathcal{P}_n$



We are bounded by how far the square extends outwards, and the tree contains the unit square so $0 < \mathcal{H}^2(\mathcal{P}) < \infty$. Thus $\dim_H \mathcal{P} = 2$

Conclusion

- Hausdorff metric describes convergence of geometric structures.
- Hausdorff measure assigns sizes to sets in any metric space, and allows us to investigate dimensions
- Self-similarity + OSC gives practical formulae for the Hausdorff dimension.
- Fractals such as Cantor, Koch, Sierpiński, Pythagoras tree have fascinating properties and some have irrational dimensions

Thank You

Please ask any questions!